

IT 520 — Which part to improve: find the bottleneck in a kitchen

In-class activity, 20 minutes. Work with the person next to you. Keep this sheet — it is today's participation record.

Tuesday nights at a restaurant, tickets that take **6 minutes** on a quiet evening take **12** during the rush. The head chef already has an answer:

"It's the oven. A tray takes six minutes and everything waits on it. Get me a second oven and this goes away."

A second oven costs money. Before anyone buys one, do what we did with the server: read the records the kitchen already keeps, work out what one ticket costs each station, find the tallest bar. **You are not timing anything** — both records were already written down.

The ticket log – Tue 18:00-19:00 (the rush)

ticket	fired	served	took
0142	18:03	18:15	12 min
0143	18:05	18:16	11 min
0144	18:07	18:20	13 min
... 12 more tickets in the hour ...			
tickets served: 15 average: 12 min			

Same log, quiet Tuesday at 17:00:
tickets served: 5 average: 6 min

The station busy-minutes report
– same window, 18:00-19:00 (60 min)

prep	busy 45 min	75%
	chopping, portioning, sauces	
oven	busy 30 min	50%
	5 trays, about 6 min a tray	
plating	busy 15 min	25%
	the pass	

Step 1 — What does one ticket cost each station?

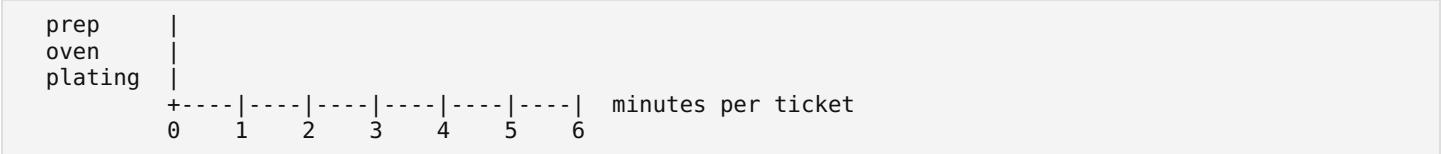
Same division as the disk this morning: **D = B ÷ C**, busy minutes of a part divided by the jobs that finished. Do it three times.

Station	Busy minutes (B)	÷ tickets served (C)	= minutes per ticket (D)
prep			
oven			
plating			

Check yourself before you go on. Add your three numbers: _____ minutes. The log says a ticket takes 6 minutes when nobody is waiting. If those don't match, one of your divisions is wrong — fix it before Step 2.

Step 2 — The tallest bar, and the ceiling

Draw the three bars, then answer.



Bottleneck (the tallest bar): _____ Is it the station the chef named? _____

Ceiling — the station has 60 minutes an hour to give, each ticket takes a bite: $60 \div \text{minutes per ticket} = \text{tickets an hour}$. Tonight the kitchen served 15.

Step 3 — Price the chef's answer

A second oven halves what the oven costs each ticket. New oven D = _____. Tallest bar now: _____. New ceiling: _____ an hour.

What happened to the ceiling, and why? _____

Step 4 — The change that costs nothing

The vegetables are chopped to order during the rush. Chop them at four o'clock instead and prep costs half as much per ticket. Nothing is bought.

New prep D = _____. Tallest bar now: _____. New ceiling: _____ an hour.

The line to take with you: the loudest station is not the bottleneck — the tallest bar is. Only a change that shrinks the tallest bar moves the ceiling, and you check it by measuring the same hour again.